(ii) The particle changes its direction when $v(t)$ changes its sign. Now.

if $0 \le t < 1$ then both $(t-1)$ and $(t-3) < 0$ and hence, $v(t) > 0$.

If $1 < t < 3$ then $(t-1) > 0$ and $(t-3) < 0$ and hence, $v(t) < 0$.

If $t > 3$ then both $(t-1)$ and $(t-3) > 0$ and hence, $v(t) > 0$.

Therefore, the particle changes its direction when $t = 1$ and $t = 3$.

(iii) The total distance travelled by the particle from time $t = 0$ to $t = 2$ is given by, $|s(0) - s(1)| + |s(1) - s(2)| = |1-5| + |5-3| = 6$ metres. ■

## 7.2.3 Related rates

A related rates problem is a problem which involves at least two changing quantities and asks you to figure out the rate at which one is changing given sufficient information on all of the others. For instance, when two vehicles drive in different directions we should be able to deduce the speed at which they are separating if we know their individual speeds and directions.

### Example 7.7

If we blow air into a balloon of spherical shape at a rate of $1000\,\text{cm}^3$ per second, at what rate the radius of the baloon changes when the radius is 7cm? Also compute the rate at which the surface area changes.

### Solution

The volume of the baloon of radius $r$ is $V = \frac{4}{3}\pi r^3$.

We are given $\frac{dV}{dt} = 1000$ and we need to find $\frac{dr}{dt}$ when $r = 7$.

Now, $\frac{dV}{dt} = 3 \times \frac{4}{3}\pi r^2 \times \frac{dr}{dt}$.

Substituting $r = 7$ and $\frac{dV}{dt} = 1000$, we get $1000 = 4\pi \times 49 \times \frac{dr}{dt}$.

Hence, $\frac{dr}{dt} = \frac{1000}{4 \times 49 \times \pi} = \frac{250}{49\pi}$.

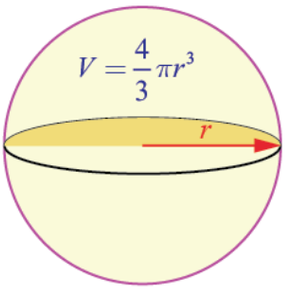


Fig.7.4

The surface area $S$ of the baloon is $S = 4\pi r^2$. Therefore, $\frac{dS}{dt} = 8\pi \times r \times \frac{dr}{dt}$.

Substituting $\frac{dr}{dt} = \frac{250}{49\pi}$ and $r = 7$, we get

$$\frac{dS}{dt} = 8\pi \times 7 \times \frac{250}{49\pi} = \frac{2000}{7}.$$

Therefore, the rate of change of radius is $\frac{250}{49\pi}$ cm/sec and the rate of change of surface area is $\frac{2000}{7}$ $\text{cm}^2/\text{sec}$. ■



Chapter 7 Differential Calculus Original-new.indd 6 26-02-2020 17:21:44